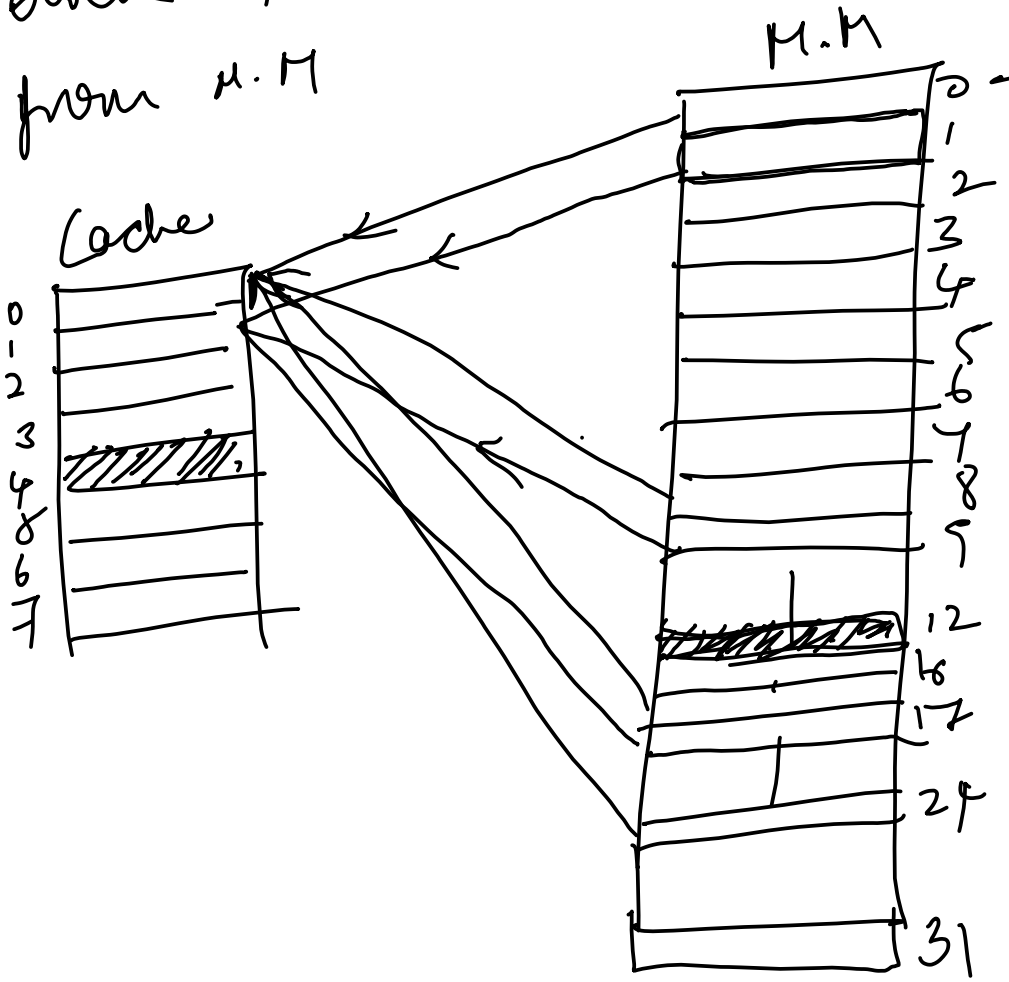
Direct mapping :-

i mod j → no. of blocks in cache

↓

block no./address from M.M



i mod j cache

M.M (0) mod 8 = 0

1 mod 8 = 1

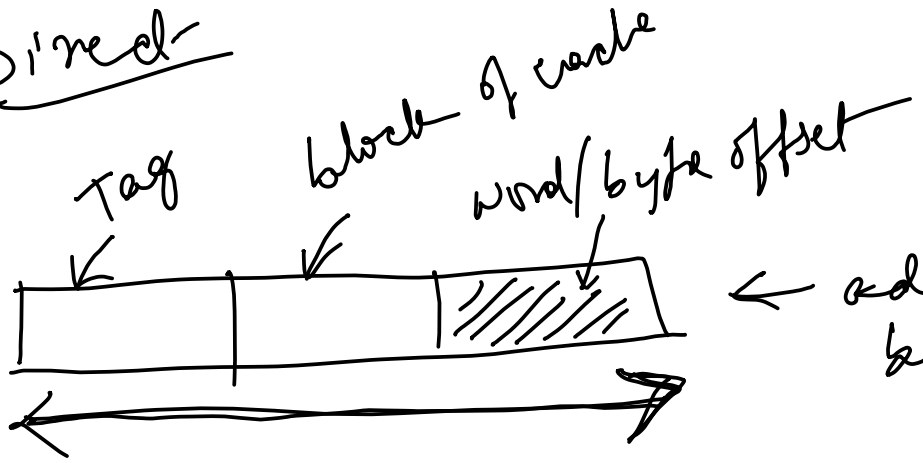
7 mod 8 = 7

8 mod 8 = 0

Cache →	0	1	2	3	4	5	6	7
M.M →	0	1	2	3	4	5	6	7
	8	9	10	11	(12)	13	14	15
	16	17	18	19	20	21	22	23
	24	25	26	27	28	29	30	31

$$\Rightarrow \frac{\text{no. of block in M.M}}{\text{no. of blocks in cache}} = \frac{32}{8} = \frac{2^5}{2^3} = 2^2$$

Direct



Cache

no. of blocks = 8

no. of bytes/block = 8

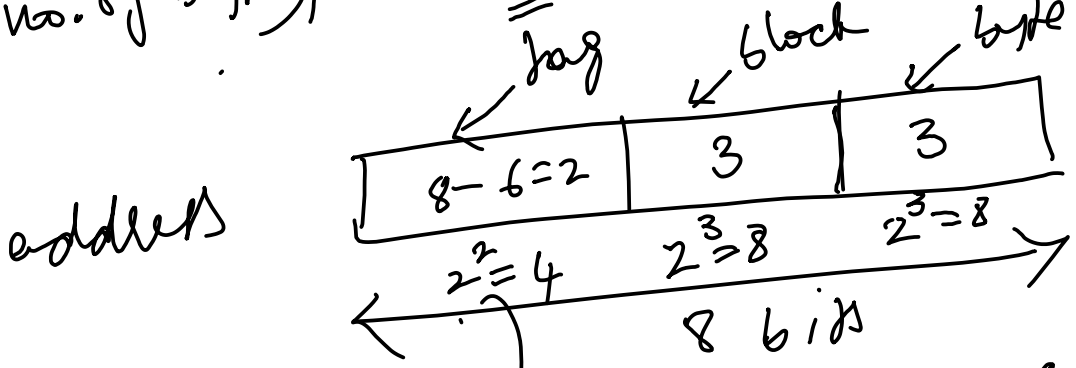
M.M

no. of blocks = 32

block size = 8

$2^3 \times 2 = 2^4$ (8)

= 256 bytes

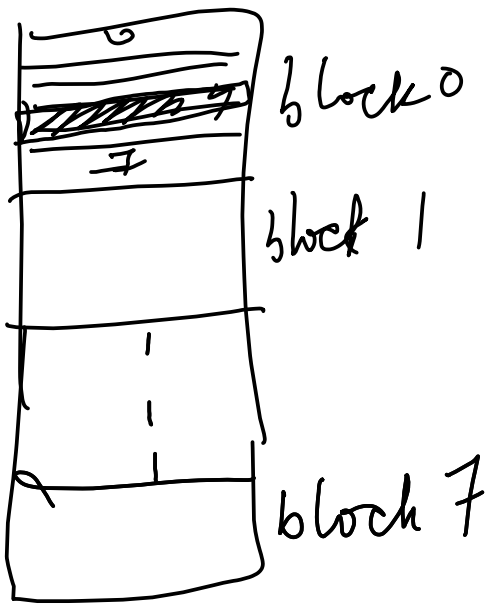


a_7 ... a_0

000...0 ⇒ 0

111...11 ⇒ 255

$\frac{32}{8} = \frac{2^5}{2^3} = 2^2$



$2^0 = 1$	$2^{10} = 1K$	$2^{20} = 1M$	$2^{30} = 1G$
$2^1 = 2$	$2^{11} = 2K$	$2^{21} = 2M$	$2^{31} = 2G$
$2^2 = 4$			
$2^3 = 8$			
$2^4 = 16$			
$2^5 = 32$ →	$2^{15} = 32K$	$2^{25} = 32M$	$2^{35} = 32G$
$2^6 = 64$			
$2^7 = 128$			
$2^8 = 256$			
$2^9 = 512$	$2^{19} = 512K$	$2^{29} = 512M$	$2^{39} = 512G$

Set-associative : n-way

↓

no. of blocks per set

2-way — 2 blocks/set

4-way — 4 blocks/set

8-way



$8/4 = 2$ sets

1-way

$8/1 = 8$ sets

8-way

$8/8 = 1$ set

i mod j

↓ ↳ no. of sets

block address from M.M

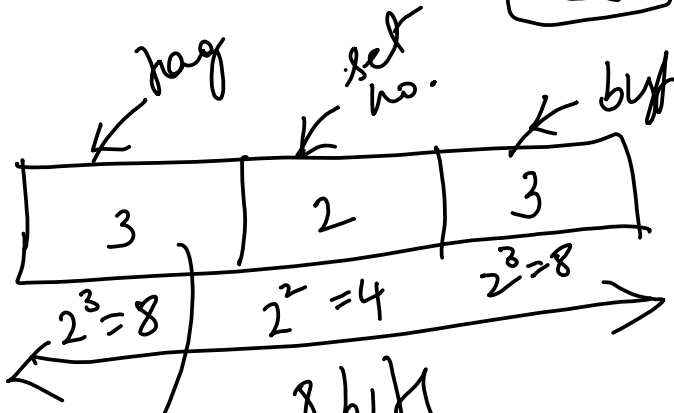
0 mod 4 = 0
1 mod 4 = 1
2 mod 4 = 2
3 mod 4 = 3
4 mod 4 = 0

Cache

set 0	set 1	set 2	set 3
0	1	2	3
4	5	6	7
8	9	10	11
12	13	14	15
16	17	18	19
20	21	22	23
24	25	26	27
28	29	30	31

M.M → blocks

2-way



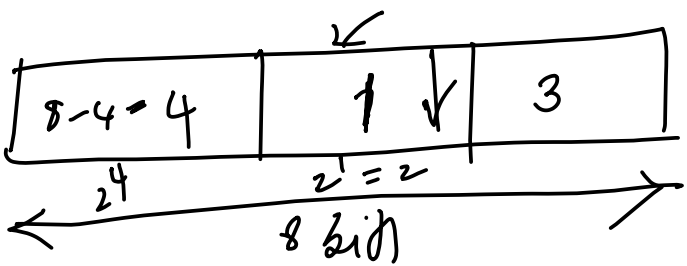
block size = 8 bits

no. of sets = $\frac{\text{no. of blocks}}{2}$

$= \frac{8}{2} = 4 = 2^2$

no. of blocks in M.M / no. of sets $= \frac{32}{4} = \frac{2^5}{2^2} = 2^3$

4-way



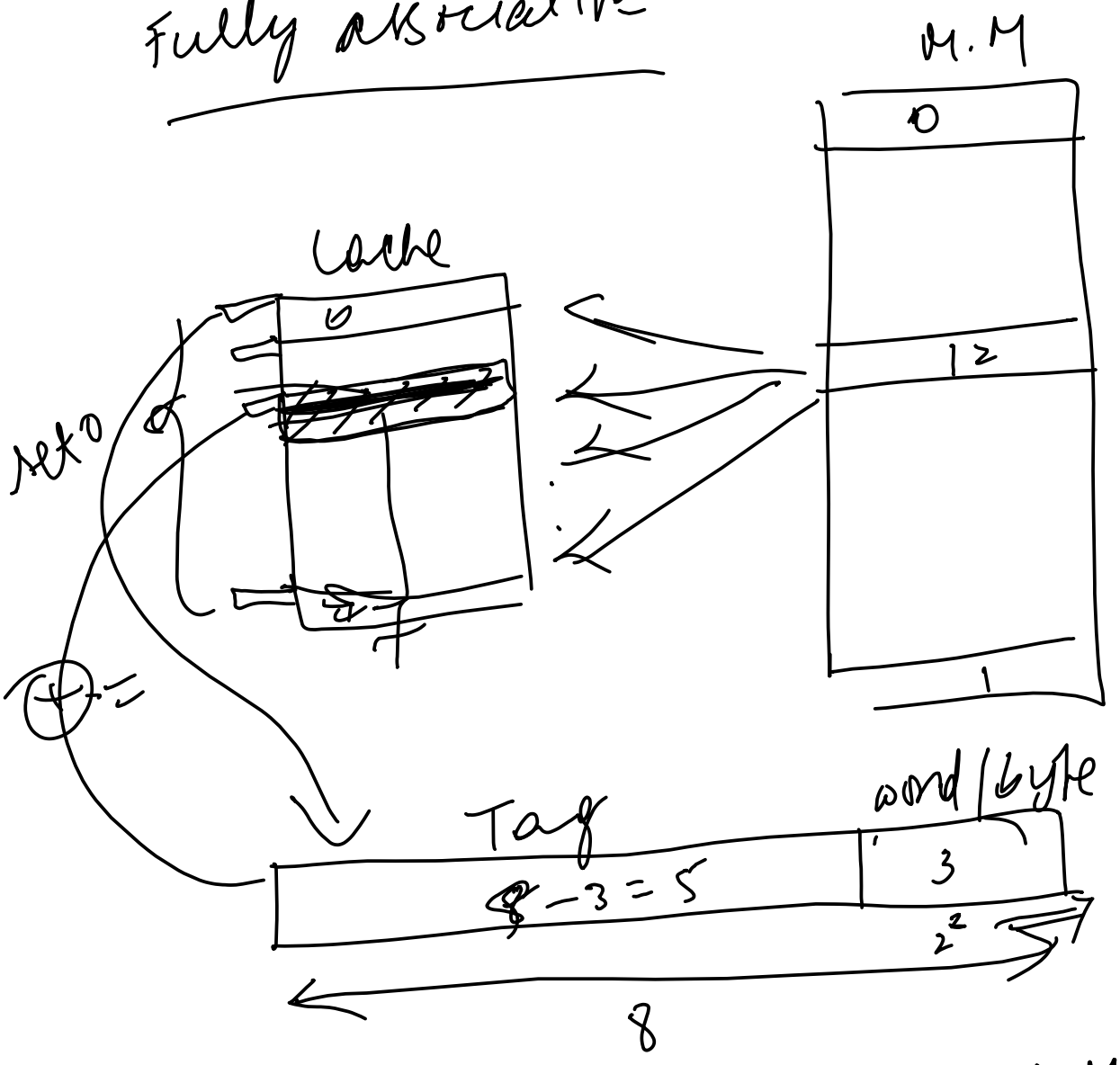
$\frac{8}{4} = 2$ sets

set 0	set 1
18	16

Fully associative = n-way set-associative

8-way

8 blocks/set



Set associative cache of size 8 blocks

- 8-way ⇒ Fully
- 1-way ⇒ Direct

2)

cache

128 blocks

16 words / block

2048 words (2K)

M.M

4K blocks

16 words / block

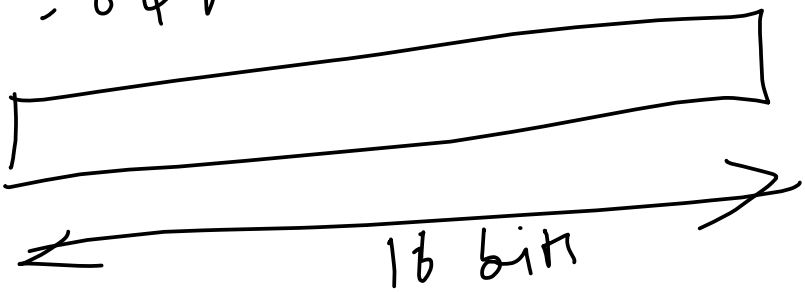
64K words

(i) calculate the bits in M.M. address

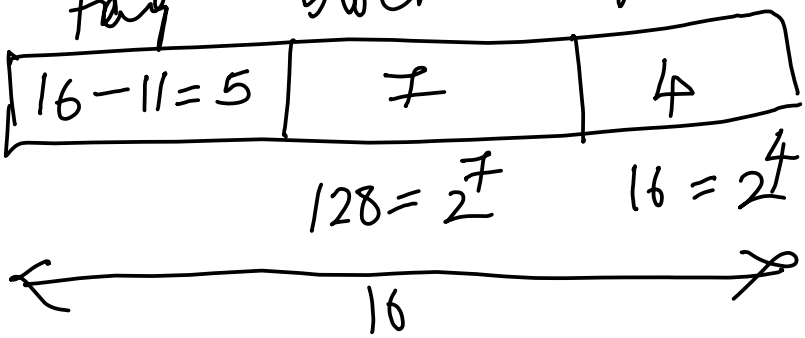
(ii) calculate the size of various fields in a) Direct b) 2-way set c) 4-way set d) fully associative

(i) M.M size in words / bytes = 64K

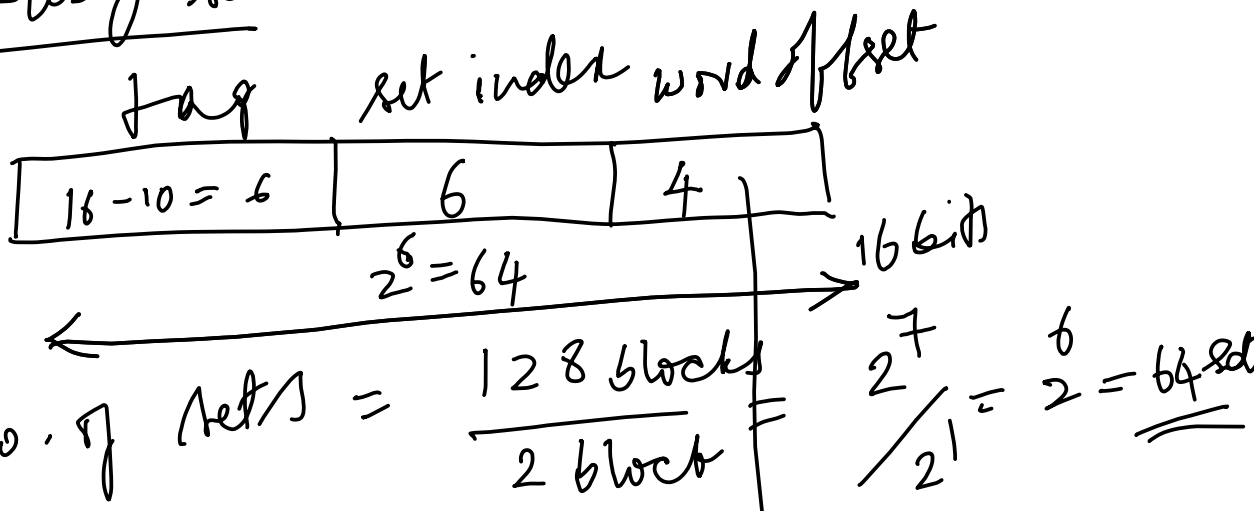
$2^{16} = 64K$



(ii) Direct

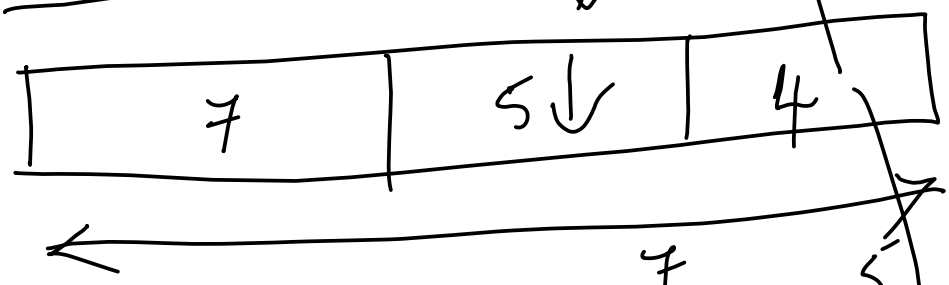


b) 2-way set

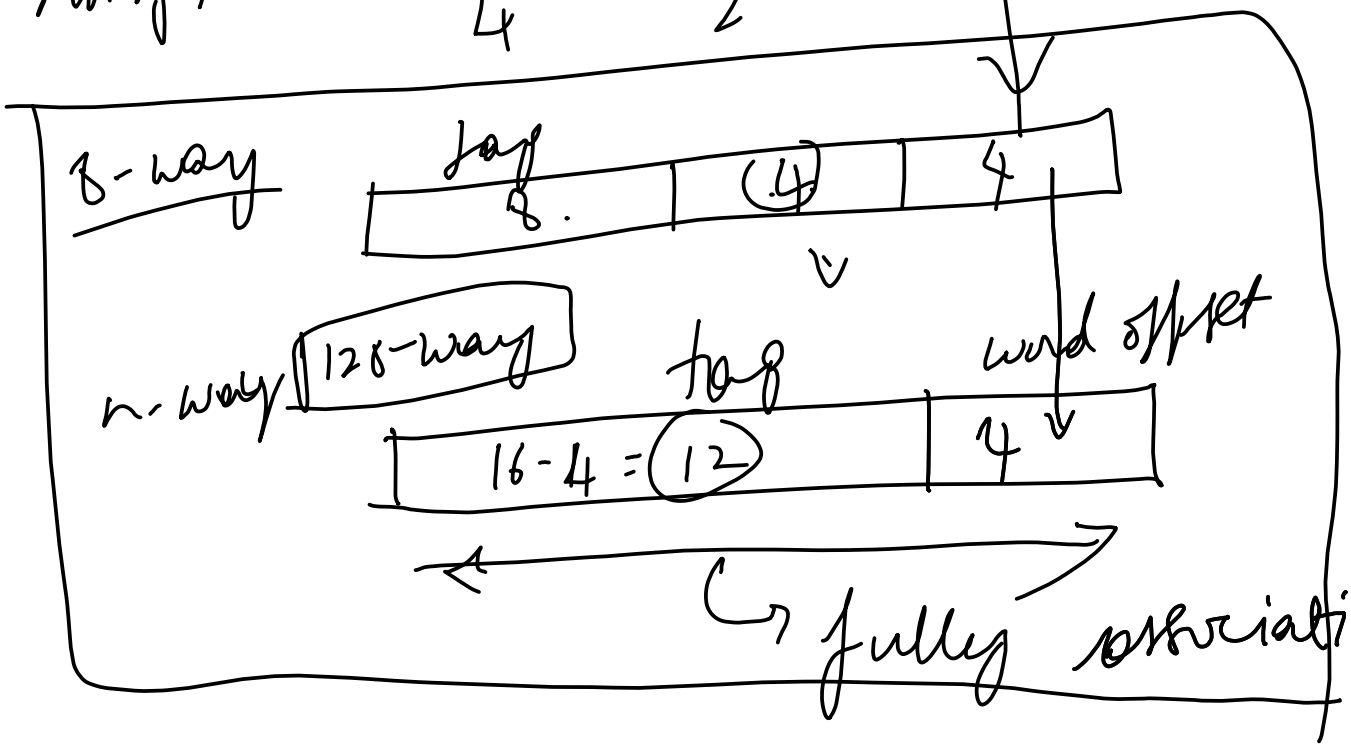


no. of sets = $\frac{128 \text{ blocks}}{2 \text{ block}} = \frac{2^7}{2^1} = 2^6 = 64$ sets

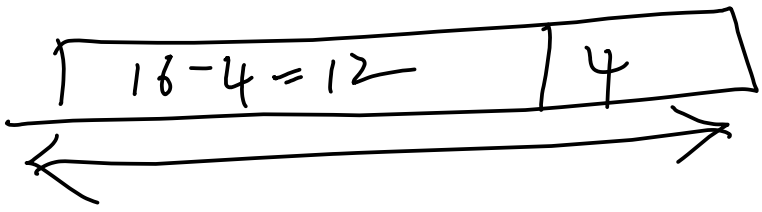
c) 4-way set ng



no. of sets: $\frac{128}{4} = \frac{2^7}{2^2} = 2^5$



d) Fully



3) A block-set-associative cache consists of a total of 64 blocks divided into 4-block sets. The M.M contains 4096 blocks each consisting of 128 words.

(a) How many bits are there in M.M address?

(b) ——k——|——n—— n — TAG, SET
k words offset ?

Soln:

M.M	cache
4096 blocks	64 blocks
128 words/block	4-way

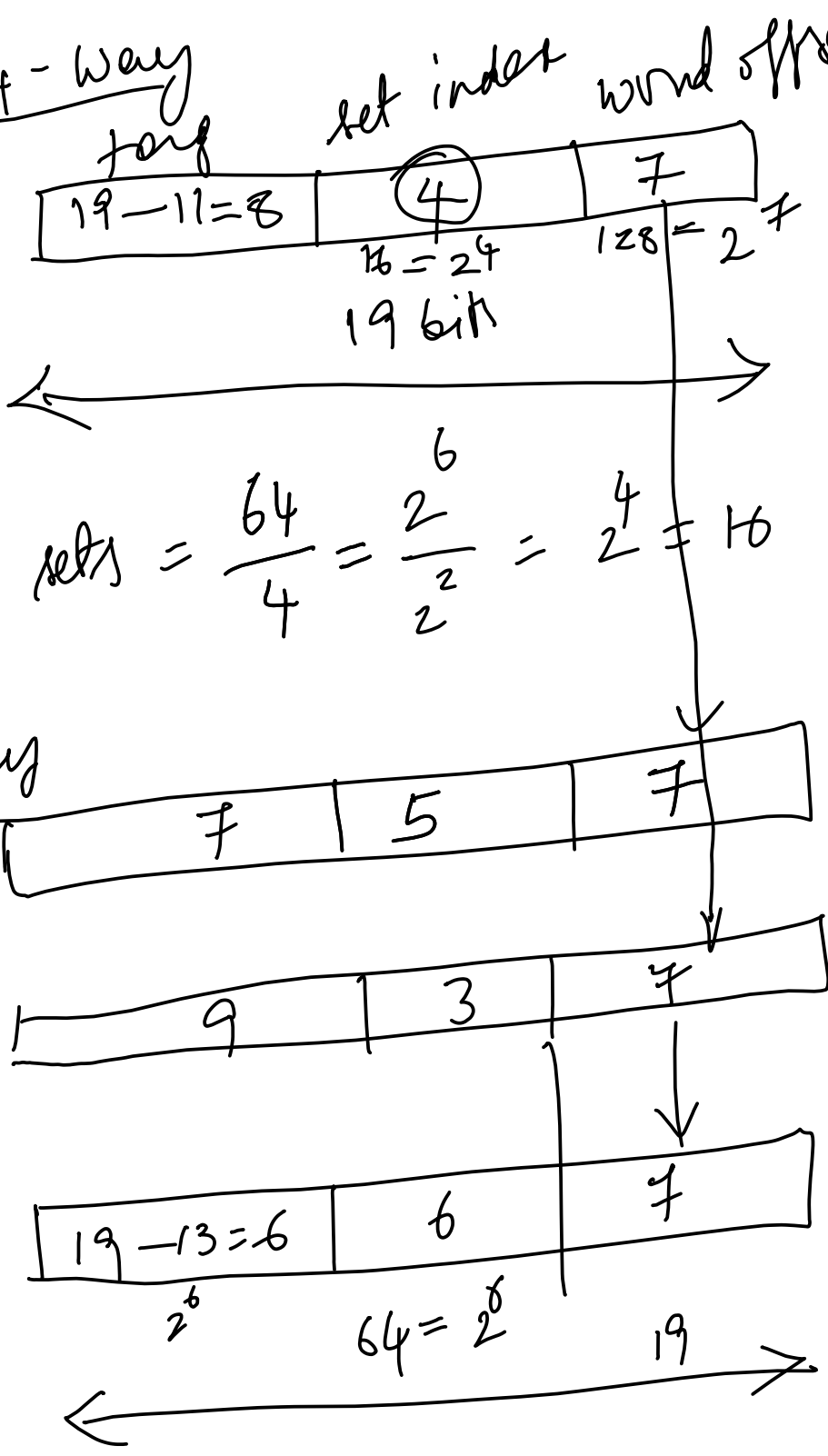
(i) Total capacity of M.M = 4096×128

$\Rightarrow 2^{12} \times 2^{7} = 2^{19}$

$\Rightarrow$ 512K

no. of bits in M.M address is 19

(ii) 4-way


tag
set index
word offset
19 − 11 = 8
4
7
16 = 2^4
128 = 2^7
19 bit
no. of sets = 64/4 = 2^6/2^2 = 2^4 = 16
2-way
7
5
7
8-way
9
3
7
Direct
19 − 13 = 6
6
7
2^6
64 = 2^6
19